

2 chapter 6

1. $f_{\max} = \frac{1}{T}$ $T > 32\text{ns}$

a) $f_{\max} = \frac{1}{32\text{ns}} \approx 31.25\text{MHz}$

b) $8\text{stages} \cdot 8\text{ns} = 64\text{ns}$

$T > 64\text{ns}$

$T = 64\text{ns}$ $f_{\max} = \frac{1}{64\text{ns}} \approx 15.625\text{MHz}$

2. (a) $A > B$

A \ B	0	1
0	0	0
1	1	0

$(A\bar{B})$

(b) $A = B$

A \ B	0	1
0	1	0
1	0	1

$(A \cdot B) + (\bar{A} \cdot \bar{B})$

(c) $A < B$

A \ B	0	1
0	0	1
1	0	0

$(\bar{A}B)$

C_{in} = 1

$$A=0, B=1, C_{in}=1$$

$$\Sigma_1 = (A \oplus B) \oplus C_{in} = (0 \oplus 1) \oplus 1 = 0$$

$$C_{out} = 1$$

$$A=0, B=1, C_{in}=1$$

$$\Sigma_2 = (0 \oplus 1) \oplus 1 = 1 \oplus 1 = 0$$

$$C_{out} = 1 \oplus 0 \oplus (A \oplus B) = 0 \oplus 1 \oplus (0 \oplus 1) = 0 \oplus 0 = 0$$

$$A=1, B=0, C_{in}=1$$

$$\Sigma_3 = (1 \oplus 0) \oplus 1 = 1 \oplus 1 = 0$$

$$C_{out} = 1$$

$$A=1, B=1, C_{in}=1$$

$$\Sigma_4 = (1 \oplus 1) \oplus 1 = 0 \oplus 1 = 1$$

$$C_{out} = 1$$

$$\Sigma_5 = AB + C_{in}(A \oplus B)$$

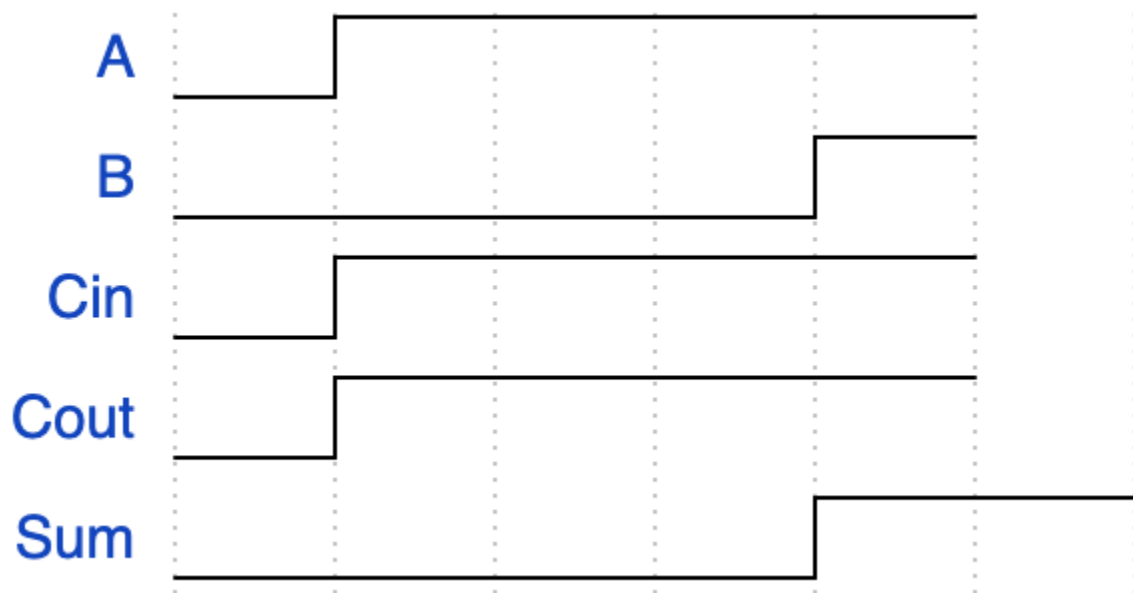
$$= 1 \cdot 1 \oplus 1(1 \oplus 1) = 1 \oplus 1(0) = 1 \oplus 0 = 1$$

$$\Sigma_1=0, \Sigma_2=0, \Sigma_3=0, \Sigma_4=1, \Sigma_5=1$$

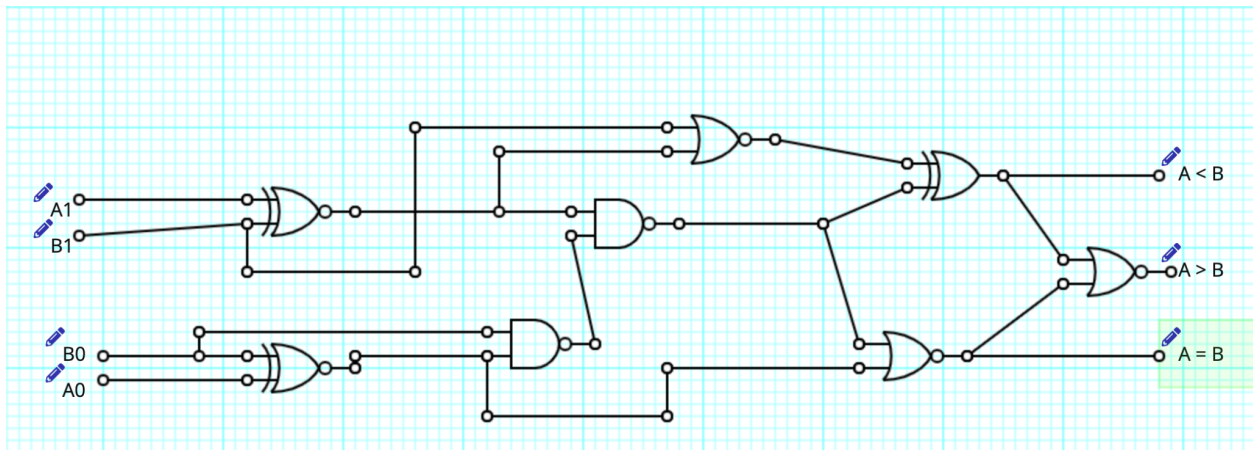
Section 2

1.

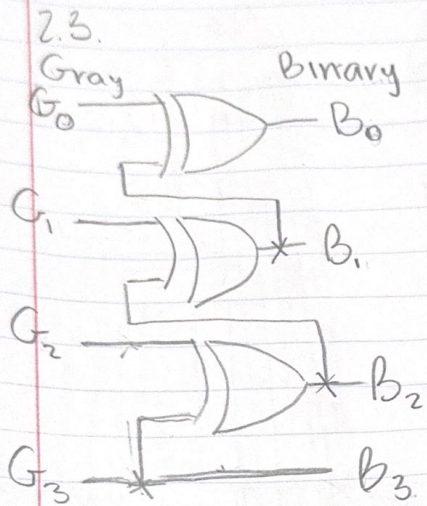
c.



2. used - https://sciencedemos.org.uk/logic_gates.php (I had trouble trying using waveDrom for this, but still wanted to make sure it was legible)



3.



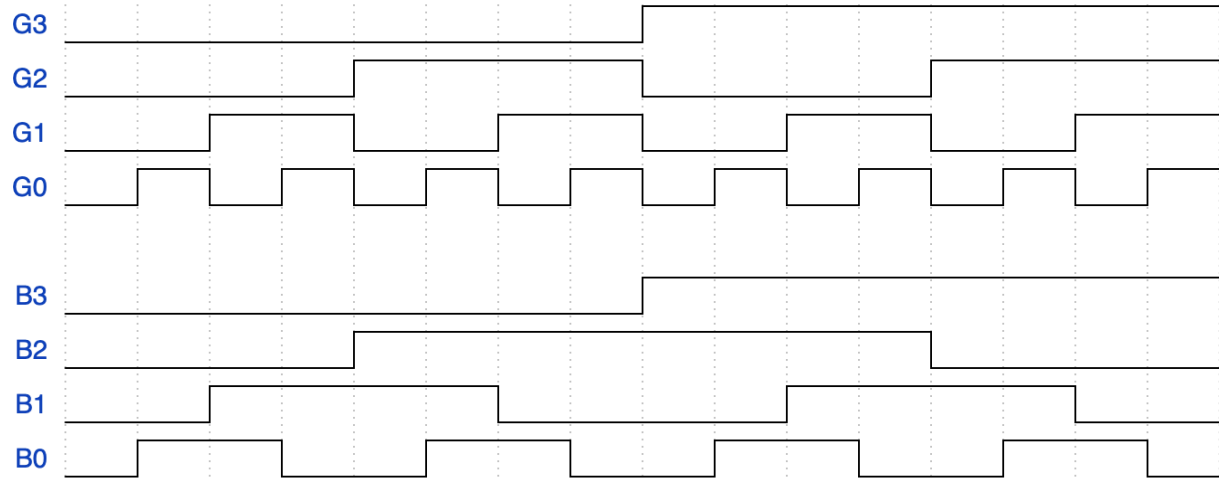
27.

$$\frac{m}{B} =$$

$$\frac{m \sqrt{2}}{h}$$

()

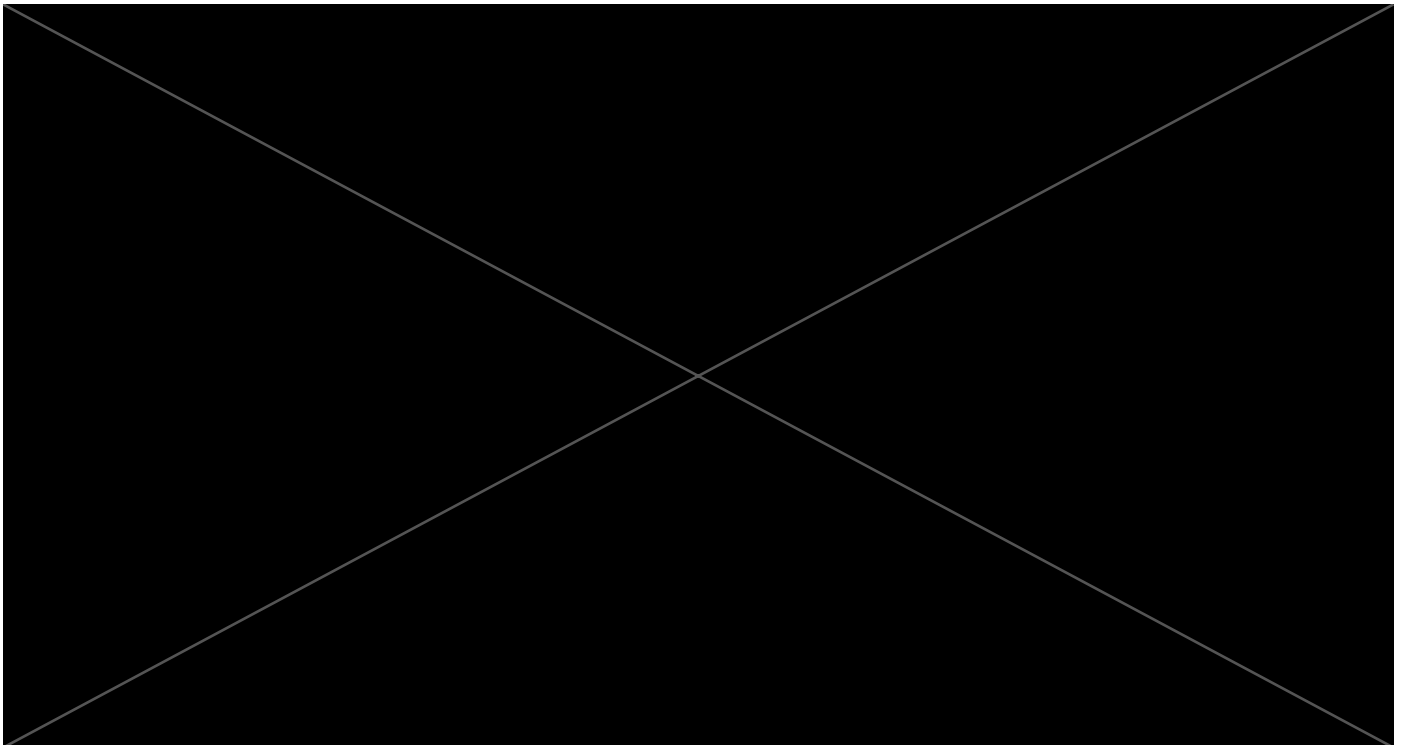
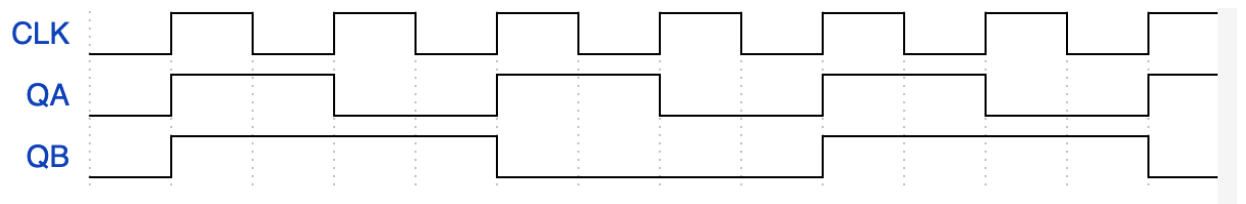
0.



Section 3

1.

d).



4. Hex to dec

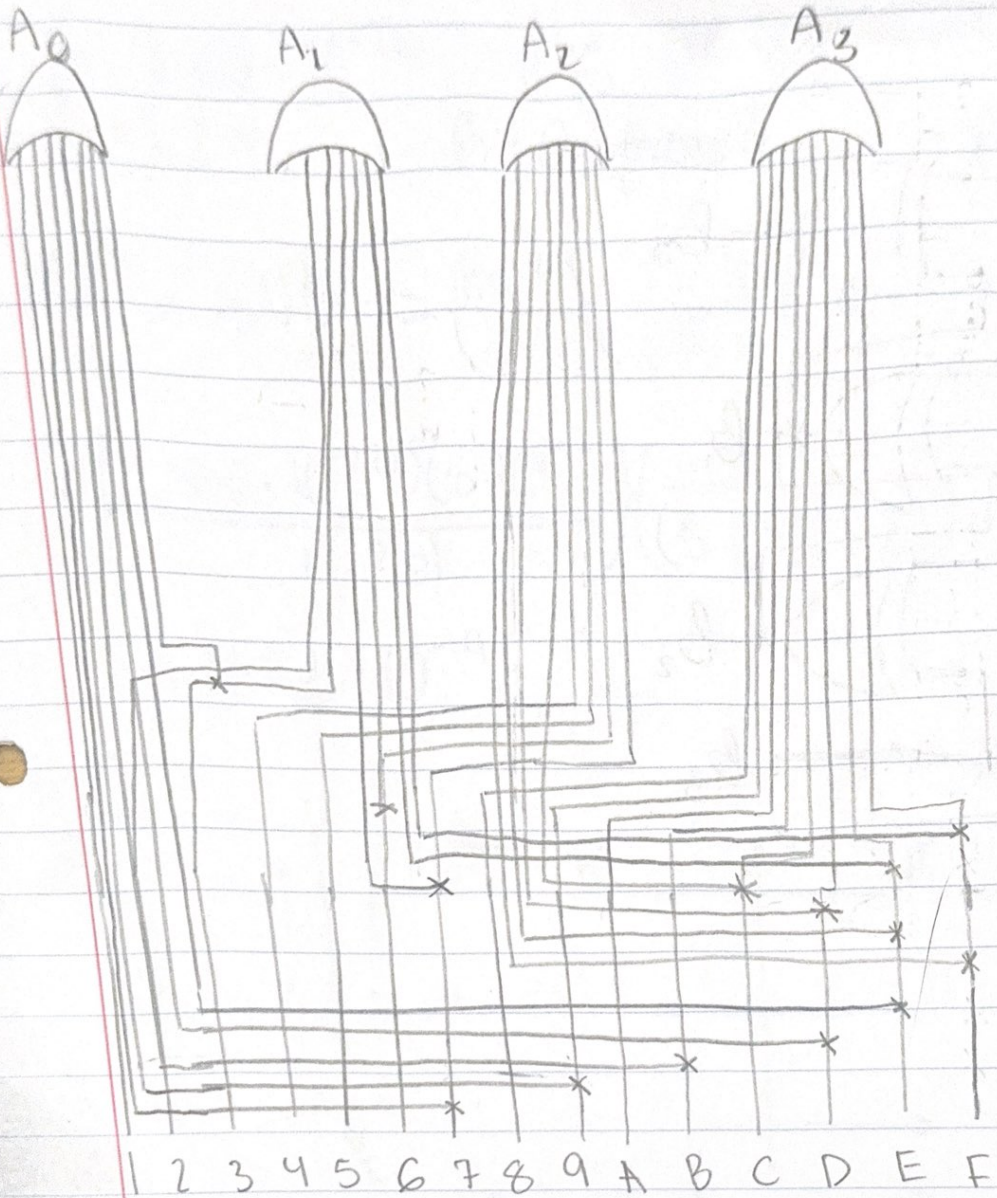
	A_3	A_2	A_1	A_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
A	1	0	1	0
B	1	0	1	1
C	1	1	0	0
D	1	1	0	1
E	1	1	1	0
F	1	1	1	1

$$A_0 = 1 + 3 + 5 + 7 + 9 + \overset{B}{11} + \overset{D}{13} + \overset{F}{15}$$

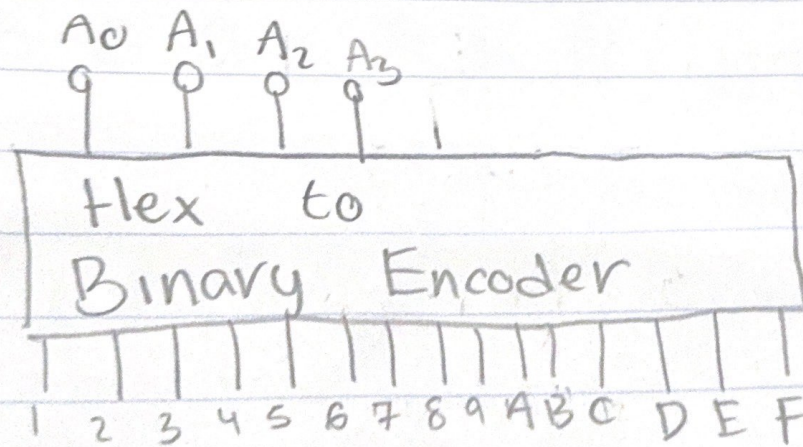
$$A_1 = 2 + 3 + 6 + 7 + 10 + 11 + 14 + 15$$

$$A_2 = 4 + 5 + 6 + 7 + 12 + 14 + 15$$

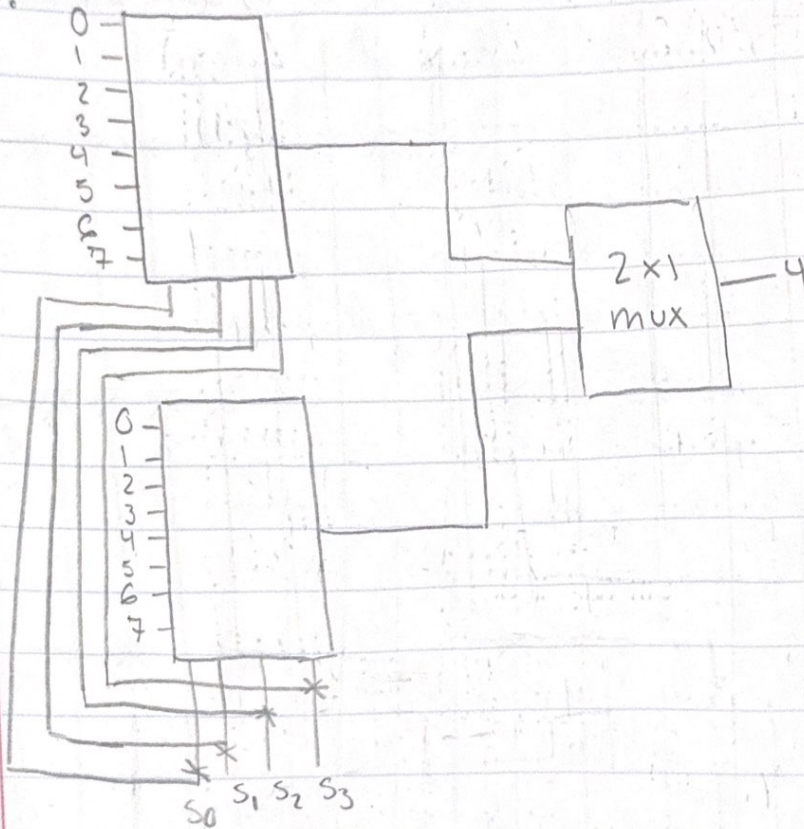
$$A_3 = 8 + 9 + 10 + 11 + 12 + 13 + 14 + 15$$

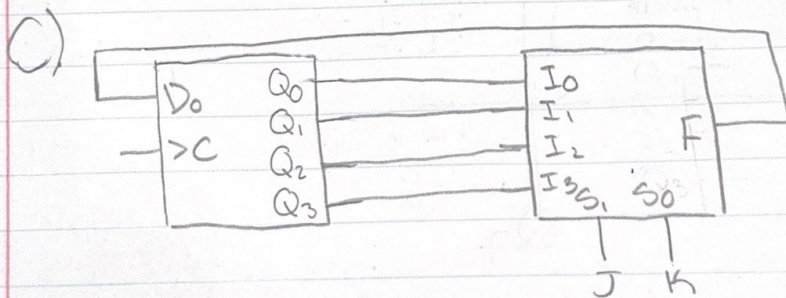
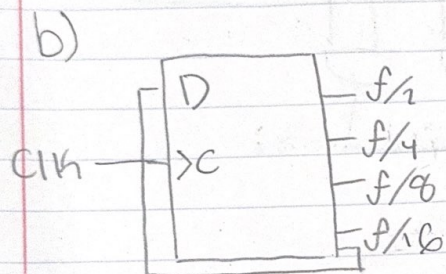
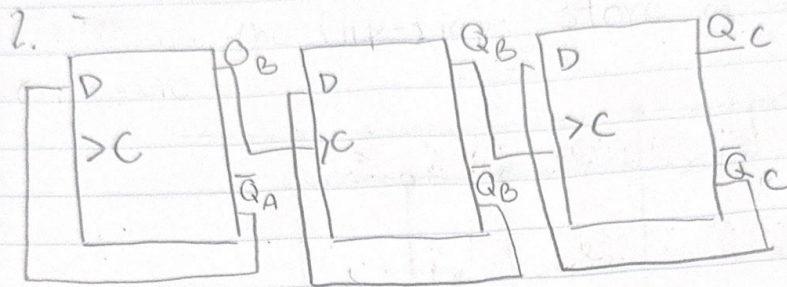


b)



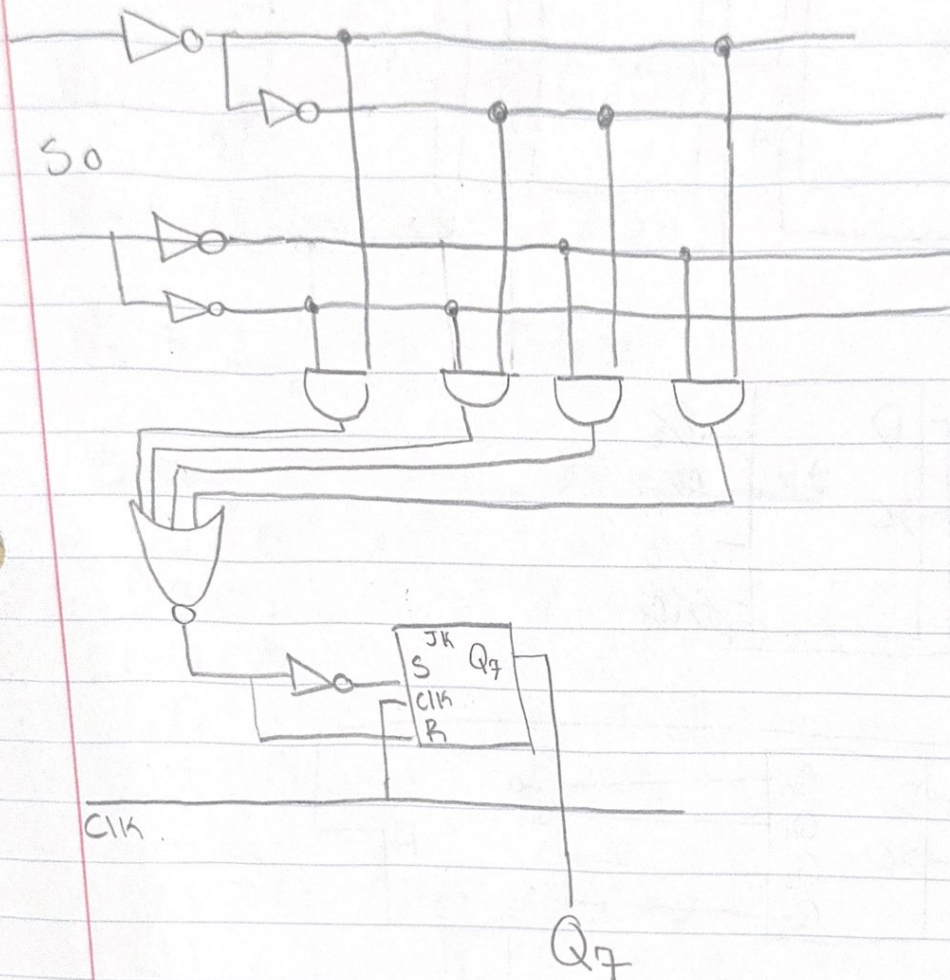
C.





Section 4

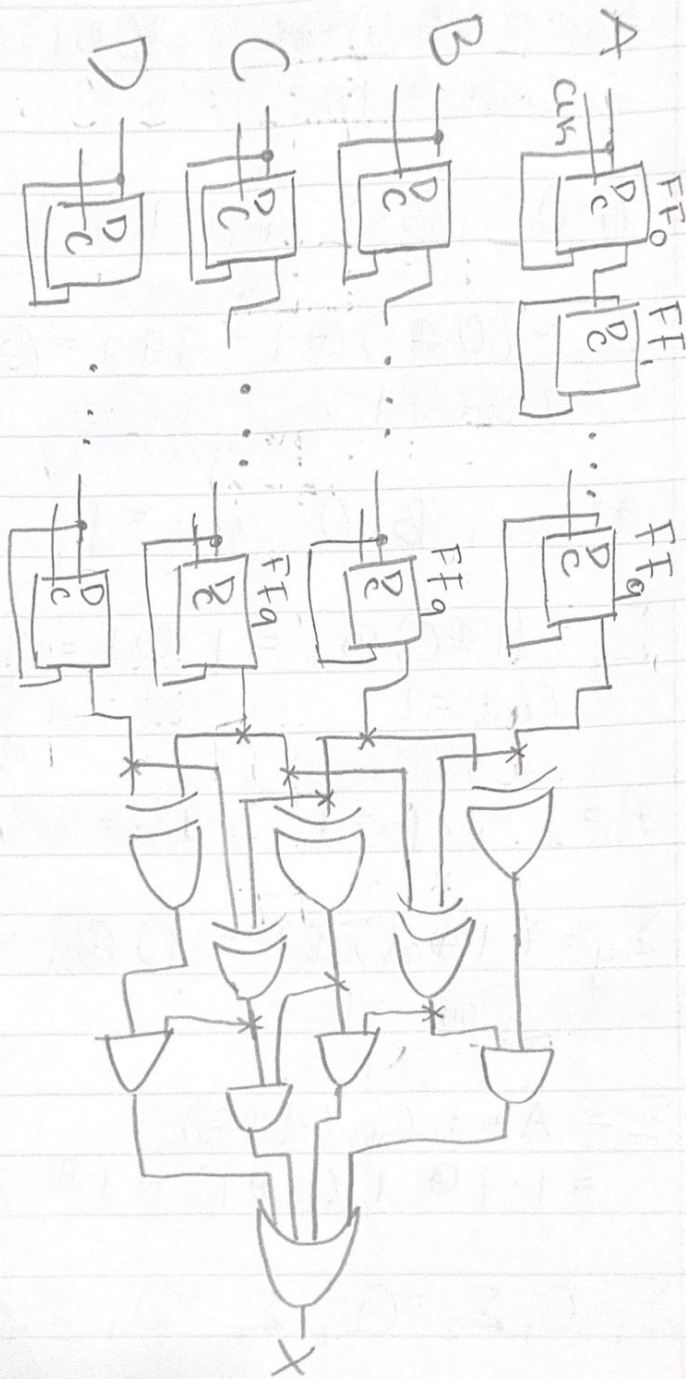
1. S₁

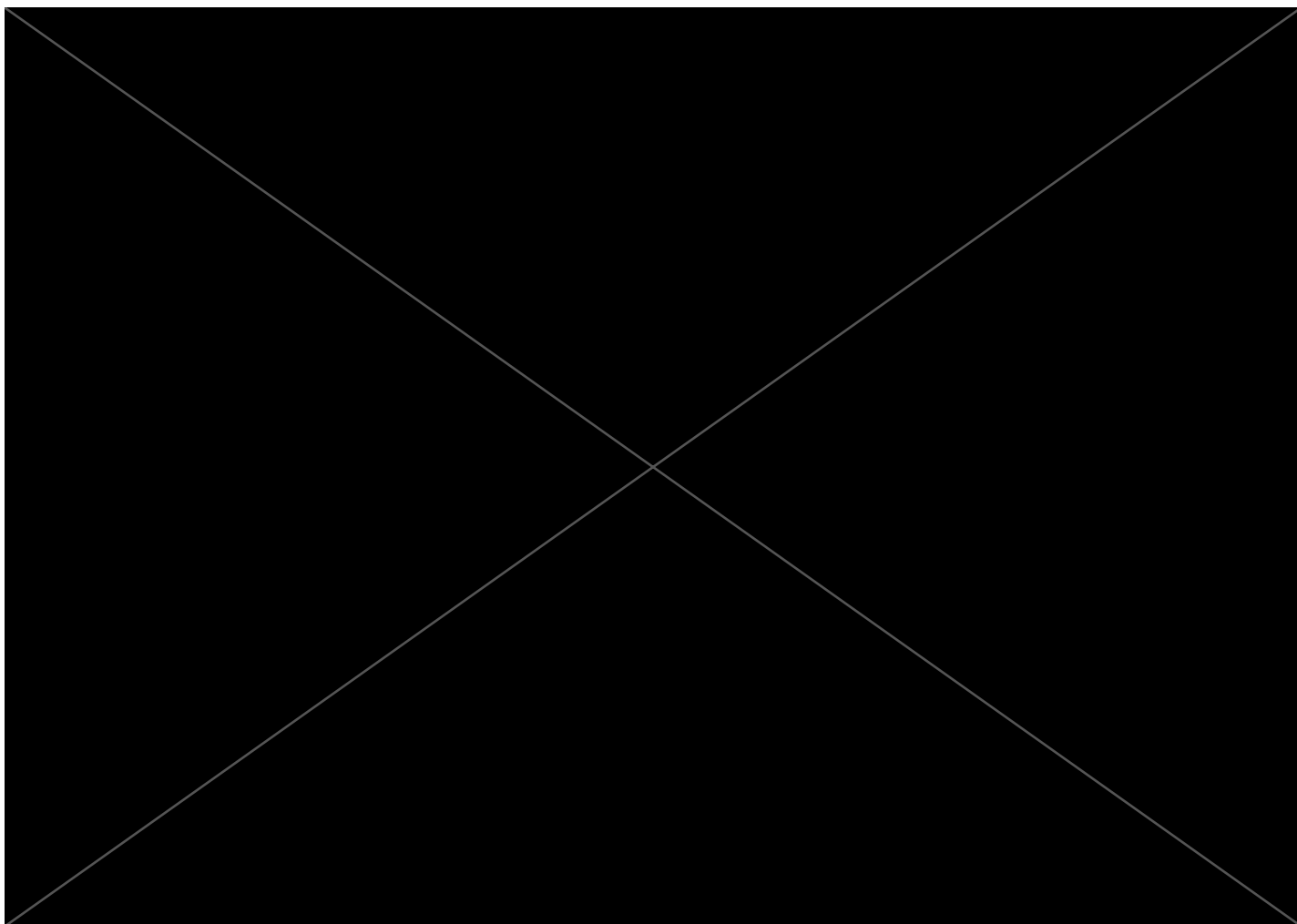


This 7 move times
for each output

4.2

$$f = \frac{1}{100 \text{ ns}} = 10 \text{ MHz}$$

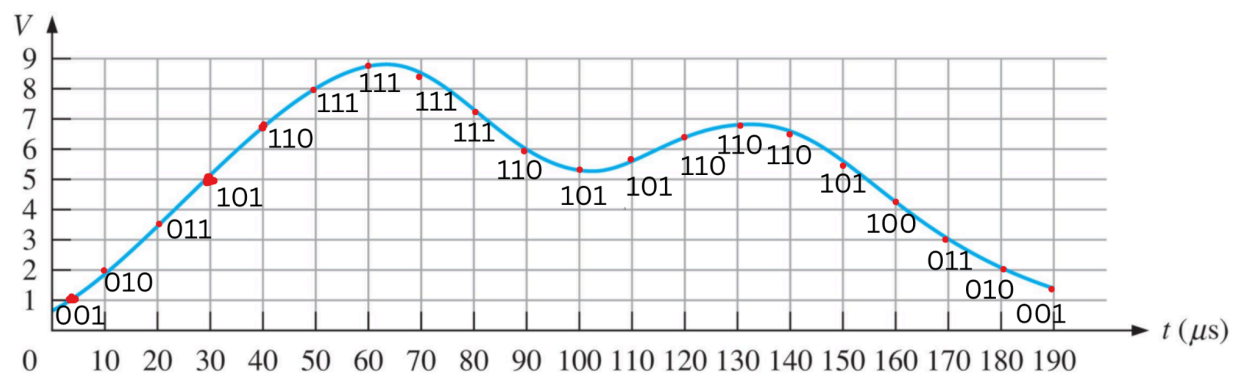




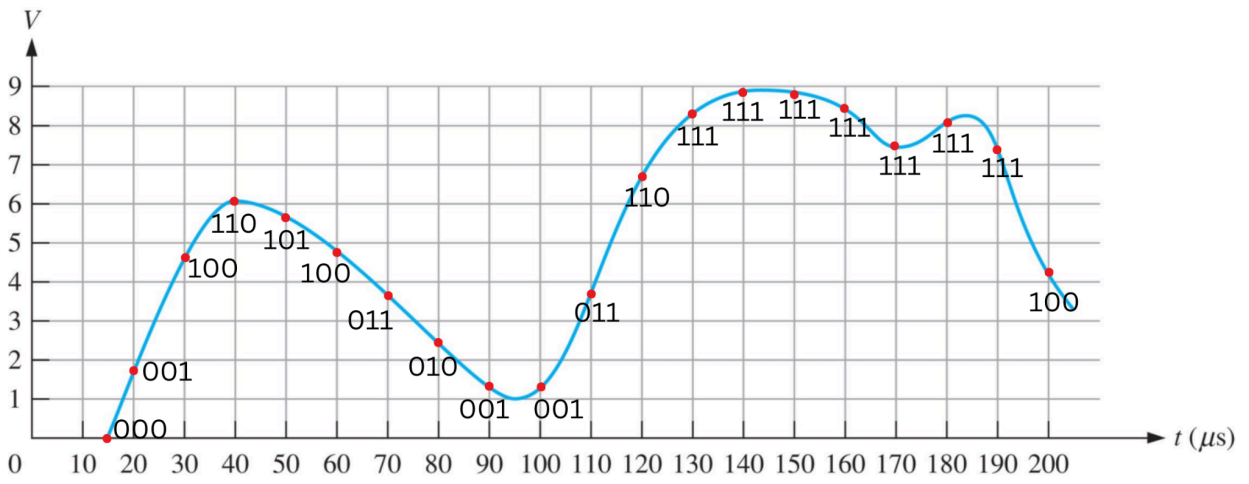
Section 5

1.

a)



b)



5.2.

$$I_0 = \frac{5V}{200k\Omega} = 0.025mA$$

$$I_1 = \frac{5V}{100k\Omega} = 0.05mA$$

$$I_2 = \frac{5V}{50k\Omega} = 0.1mA$$

$$I_3 = \frac{5V}{25k\Omega} = 0.2mA$$

$$V_{out} = D_0 = (10k\Omega)(-0.025mA) = -0.25V$$

$$V_{out} = D_1 = (10k\Omega)(-0.05mA) = -0.5V$$

$$V_{out} = D_2 = (10k\Omega)(-0.1mA) = -1V$$

$$V_{out} = D_3 = (10k\Omega)(-0.2mA) = -2V$$

